MANCHESTER 1824
The University of Manchester

Comparing Arm Tracking Teleoperation to Conventional Joystick Control of a Laterally Mounted Robot Arm

A dissertation submitted to The University of Manchester for the degree of

Bachelor of Engineering in Electronic Engineering

in the Faculty of Science and Engineering

Year of submission

2026

Student ID

11006231

School of Engineering

Contents

Contents 2

Abstract 4

Declaration of originality 5

Copyright statement 6

Acknowledgments 7

1 Introduction 8

1.1 Background and motivation 8

1.2 Aims and objectives 9

2 Examination of existing human to robot arm teleoperation methods 9

2.1 Why choose teleoperation over conventional input methods 9

2.2 Requirements of a suitable tracking system 10

2.3 Survey of existing tracking technologies 11

3 Comparing two viable computer vision based methods of robot arm teleoperation 13

3.1 Comparison of both computer vision methods 13

3.2 Results of joint angle tracking methods 16

3.3 Final Tracking Method 18

4 Arm tracking vs conventional control experiments 18

4.1 Description of experiment to compare conventional vs human arm tracking 18

4.2 Scenario 1: Random position matching task 19

4.3 Scenario 2: Simple environment pick and place task 19

4.4 Scenario 3: Obstacle-rich environment pick and place task 21

5 Conclusions, future work and reflections 22

5.1 Conclusions 23

5.2 Future work 23

5.3 Project Management Reflections 24

References 24

Appendices 27

A Project outline **27**
A.1 Overview and motivations 27
A.2 Deliverables 27

B Risk assessment **29**

C Code **30**

D DH Parameters **30**

E ANOVA Figures **30**

Word count: 4395

Abstract

Robot arms have been engineered to mimic the dexterity of human arms, yet conventional joystick control methods do not make full use of this. Prior research has been undertaken to explore the performance impact of mapping the end effector of the robot arm to the operator's hand, with demonstrated performance benefits. This project aims to explore whether full mapping of an operator's arm movements to control an industrial robot with laterally mounted arms (YuMi) results in similar performance benefits.

Secondary research is performed to sort viable approaches, two of which are found to be equally plausible: pose estimation with a neural network (MediaPipe) as well as computer vision based marker tracking (ArUco). To distinguish these two methods, a minimum viable version of each is implemented and then an experiment is devised to find the most suitable method. The ArUco markers showed a low detection rate, whereas pose estimation had some depth-axis related distortion but was able to show sufficient angle tracking accuracy with 3D movements. Hence, the pose estimation approach is further developed to interface with a robot simulation.

In practice, the joystick approach is 42.1% faster on average on a task that involves moving the end effector to a random position in the workspace. Pose estimation is faster than joystick control by 24.4% in a scenario involving grabbing an object in one location and moving it to a target destination. In a similar scenario maneuvering in an obstacle field, the pose estimation solution is 26.7% faster and resulted in 3.2 fewer collisions on average.

These findings are due to the higher information density and intuition of the pose estimation control method - and limitations of typical inverse kinematic solvers hindering the joystick method. Likewise, the results are from a sample size of 1 operator, and therefore are not fully representative of generalised performance benefits.

Declaration of originality

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Copyright statement

i The author of this thesis (including any appendices and/or schedules to this thesis) owns certain copyright or related rights in it (the "Copyright") and s/he has given The University of Manchester certain rights to use such Copyright, including for administrative purposes.

ii Copies of this thesis, either in full or in extracts and whether in hard or electronic copy, may be made *only* in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has from time to time. This page must form part of any such copies made.

iii The ownership of certain Copyright, patents, designs, trademarks and other intellectual property (the "Intellectual Property") and any reproductions of copyright works in the thesis, for example graphs and tables ("Reproductions"), which may be described in this thesis, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.

iv Further information on the conditions under which disclosure, publication and commercialisation of this thesis, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy (see `http://documents.manchester.ac.uk/DocuInfo.aspx?DocID=24420`), in any relevant Thesis restriction declarations deposited in the University Library, The University Library's regulations (see `http://www.library.manchester.ac.uk/about/regulations/`) and in The University's policy on Presentation of Theses.

Acknowledgments

Thank you to Dr. Simon Watson who was able to provide invaluable feedback and guidance throughout the project. Thanks also to PepsiCo for providing the caffeine for me to get through many an 'early' 10:00 A.M. meeting with him. Gratitude is also directed towards Mike Oldfield for providing the soundtrack to many coding-laden evenings such that my mind would not have to suffer the silence of a cold Mancunian evening. Finally, an uncountable amount of thanks goes to my father for granting me the world and more.

1 Introduction

The goal of this project is to create a system for controlling and teleoperating a robot arm with a human arm. In order to maintain simplicity in the project scope, the robot arm is assumed to be shoulder-mounted and has the seven degrees of freedom (DoF) [1], matching that of a healthy human arm. An example of the humanoid robot arm (the YuMi robot) to be used for this project is shown in Fig. 1.

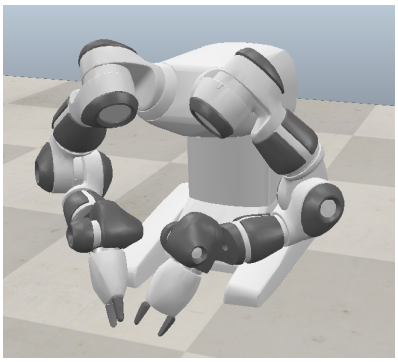


Fig. 1. YuMi Robot Arm To Be Used In Experiment.

1.1 Background and motivation

The motivation for this project comes from articles within the books: “Human–robot teleoperation based on Imus” [2] and “Whole-body human-motion based robot teleoperation” [3]. Both of these articles use an inertial measurement unit (IMU) to gather data, but only maps the wrist position to the robot’s end effector.

Redundancy in the degrees of freedom of the arm means that a valid end effector position can have multiple positions of the previous links [4]. Therefore this ambiguity could lead to collisions with obstacles in the environment, and lack of safety for people within the envelope of the robot arm.

Some specific use cases will require absolute control of every link of the robot arm - for example underwater sample collection. Whilst Jin et al. describes a potential autonomous solution [5], this method is comparatively slow to the prospect of remote teleoperation. Consequently, the complexity increase for the case where nothing is stationary, a more realistic use case, would mean that this method would be inefficient compared to the intuition of a trained operator.

Additionally, Jankowski and Grabowski shows that for tasks involving teleoperation of a robot for

pick and place tasks, virtual reality style mapping of head movements with arm to end effector mapping is more efficient and less error prone than a typical joystick [6]. Hence, exploration of the performance characteristics between conventional control methods and arm mapped teleoperation is undertaken.

1.2 Aims and objectives

The overall aim is to compare the performance of a system for mapping the joint angles of a human arm to a shoulder-mounted robot arm to conventional control methods. This will include:

- Identifying and comparing existing technologies, including IMU based solutions discussed earlier, attitude and heading reference systems [7], computer vision based systems [8], and flex sensor based systems [9].
- Developing a version of the tracking system suitable for controlling a simulation of the YuMi robot arm.
- Devising experiments appropriate to comparing conventional methods of control to arm tracking teleoperation methods.
- Analysing the difference, if any, in performance between either method of control in a variety of environments.

2 Examination of existing human to robot arm teleoperation methods

This section explores which arm tracking methods are the most suitable for this project. Quantitative and qualitative data from secondary research is used to create a justified specification for the final solution. Finally, a Pugh matrix will be applied to fairly score and select an appropriate method to further develop.

2.1 Why choose teleoperation over conventional input methods

The measurement of human arm joint angles has been an interest of academia for several decades: with one of the first examples being demonstrated by Liu et al., [10]. In industrial settings, full body

tracking has primarily been used in the film industry to accurately measure an actor's performance [11] and map them onto computer generated characters [12].

This approach could also be applied to the control of robotics: have a skilled operator remotely control industrial robots where traditional control methods (e.g. joysticks) are unsuitable. Research has shown that this approach could lead to 1.8 times faster task completion time and with a quarter of the errors (1.54% compared to 8.65%) in the best cases [6].

Another way to increase the performance of a task with teleoperation is mapping the head movements of an operator to the camera of the robot performing the task [13]. However, this study was based primarily on the control of wheeled robots; manual joystick controls exceeded the performance of teleoperation in this specific task.

These two cases suggests that whether teleoperation is more effective than traditional joysticks is strongly dependent on the task. Overall, in the case of robot arm control, there is potential for performance increases via human arm teleoperation [2], strengthening the justification for this project.

2.2 Requirements of a suitable tracking system

In order to compare the final system to existing systems and research, desired characteristics must be identified.

Accuracy requirements - Be able to measure joint rotations with a minimum resolution of $3°$.

Experienced surgeons have been measured to have $5°$ of vertical movement range on all joints when asked to hold their hand on a target, and close to $3°$ range for horizontal movement [14]. Therefore any measurement lower than this resolution can be characterised as 'noise' from human inaccuracy, and can be smoothed out.

Latency requirements - Have a maximum system latency of 300 ms.

Teleoperation task performance generally degrades around the two key figures of 600 ms and 1500 ms [15]. Thus, the Nyquist criterion equivalent of this threshold is used as a well informed estimate of how much latency can be present in the system, leaving a final requirement of 300 ms.

Resource requirements - The budget for this project is £180, and 30 unit credits worth of time, equating to 240 hours of time commitment.

Both of these factors will mean that the final solution favours simpler, more complete solutions. Anything novel is a potential failure point that could mean the project does not complete.

2.3 Survey of existing tracking technologies

In order to fully evaluate the myriad of arm tracking methods available, a Pugh Matrix has been created (see Table 1) to evaluate and compare several factors between each method. Each method will score $1 - 5$ in each category, with a brief comment justifying the score in each column.

Below are the decision factors of the Pugh Matrix and a brief description of how they are scored:

Complexity - This is calculated based on a well reasoned estimate of the total time to build the system. Each multiple of 48 hours reduces the score by 1, e.g. $0 - 48$ hours would score 5, $49 - 96$ hours would score 4.

Cheapness - This is calculated based on a well reasoned estimate of the total cost of building the system. Each multiple of £36 reduces the score by 1, e.g. £$0 -$ £36 would score 5, £$37 -$ £72 would score 4.

Accuracy - This is calculated from available research on the accuracy of this method. Each multiple of $5°$ above $5°$ reduces the score by 1, e.g. an absolute accuracy of $6 - 10°$ would score 4, $11 - 15°$ would score 3.

Latency - This is calculated from available research on the time it takes to obtain a single measurement of all 7 joint angles from this method. Each multiple of 100 ms above 300 ms reduces the score by 1, e.g. a latency of 301 ms $- 400$ ms would score 4.

Analysis of existing methods shows that the two computer vision approaches score similarly in terms of viability, therefore further analysis will be required to confirm the viability of each method. These will be performed in the form of accuracy and latency experiments. This has the advantage of being able to examine more factors than existing literature - of particular interest is the accuracy of measuring multiple joints angles simultaneously.

In particular, the potential accuracy advantage of the marker tracking method makes it more appealing, though this is offset by the stupendous cost of conventional Vicon camera systems. However, open source software along with ArUco markers could be used to offset this cost - which is worth exploring before committing to any substantial development. For example, Wieczorek et al. describes using ArUco markers along with standard home cameras to perform specialised kinematic analysis [16].

Table 1. Pugh Matrix with detailed scoring justifications

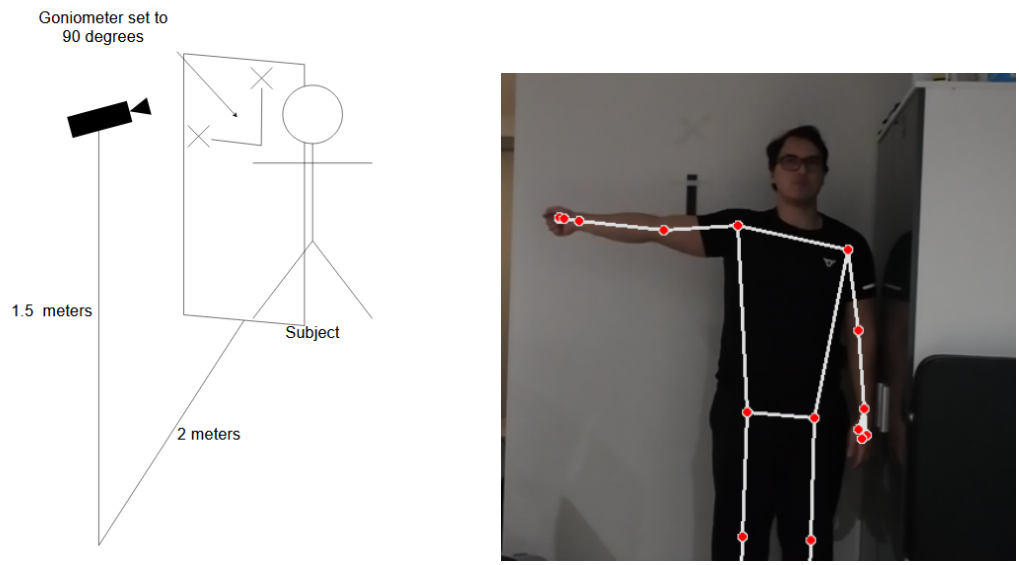
Method	**Complexity** $1-5$	**Cheapness** $1-5$	**Accuracy** $1-5$	**Latency** $(1-5)$	**Total**
Pose estimation via computer vision	5 Existing solutions reduce complexity $30-40$ hours	5 £30 USB Camera	3 $10^\circ - 15^\circ$ [17]	5 ≤ 100 ms [18]	18
Marker tracking via computer vision	5 Very complex, though can be simplified through FOSS	3 £180+ for Vicon, cheaper solutions exist	5 <5° error	5 100-200ms	18
IMU-based	3 100 hours including PCB design + drift correction	4 £$50 -$ £60	3 2.8° [19]	5 <1 ms [20]	15
Flex Sensors	2 150 hours	4 £40-60	4 10° [21]	5 ≤ 100 ms	15

3 Comparing two viable computer vision based methods of robot arm teleoperation

The criteria selected both ArUco marker tracking and pose estimation as equally viable methods of human arm tracking. This section aims to develop a minimum viable product (MVP) version of both methods to be used to quantify which approach is more suitable for the final solution.

3.1 Comparison of both computer vision methods

In order to evaluate the accuracy of both computer vision methods, a simple experiment was set up to measure the absolute accuracy on the joint angles that can be measured. A goniometer is attached to the wall at $90°$, with two end positions marked at $90°$ and $180°$. The subject will move their arm repeatedly between these markers, and the recording is captured by a camera as described in Fig. 2.



(a) A diagram of the experimental setup. **(b)** Example data capture

Fig. 2. The experimental setup, showing data capture.

The subject will then flex and extend their elbow between the two endpoints. In order to obtain a sufficient amount of data, 5 videos are recorded, each showing 10 flexion and extensions.

This creates enough samples to perform ANOVA between each video, to maintain experimental integrity. In order to measure error, the graph of elbow flexion angles is extended towards the $90°$

and $180°$ endpoints. This secondary graph is then subtracted from the first graph to obtain the error value, from which we can calculate the root mean square error, and is shown in Fig. 3.

Fig. 3. Showing how RMSE is calculated from elbow flexion data.

Upon analysis of the RMSE, it was found that this value was $25.83°$. which was greater than the literature value of $10.37°$ [17]. Frame by frame analysis of the video showed that this descrepancy was not attributable to human error - the arm was clearly flexing and extending to the correct endpoints.

Plotting the inferred joint positions frame by frame revealed the culprit - perspective! The machine learning model used for joint position extraction is finding extra movement in the depth axis of the video, and attributing it to roll in the elbow axis, instead of flexion. This is clearly shown in Fig. 4.

To analyse the performance of joint angle tracking in the three dimensional case, a simple movement path is created. A 7 DoF manipulator is created from relevant DH parameters (specifics found in appendix Table 3) These movements are then fed through a forward kinematics processing program to obtain a visualisation of this movement path, and this movement is recreated by the subject.

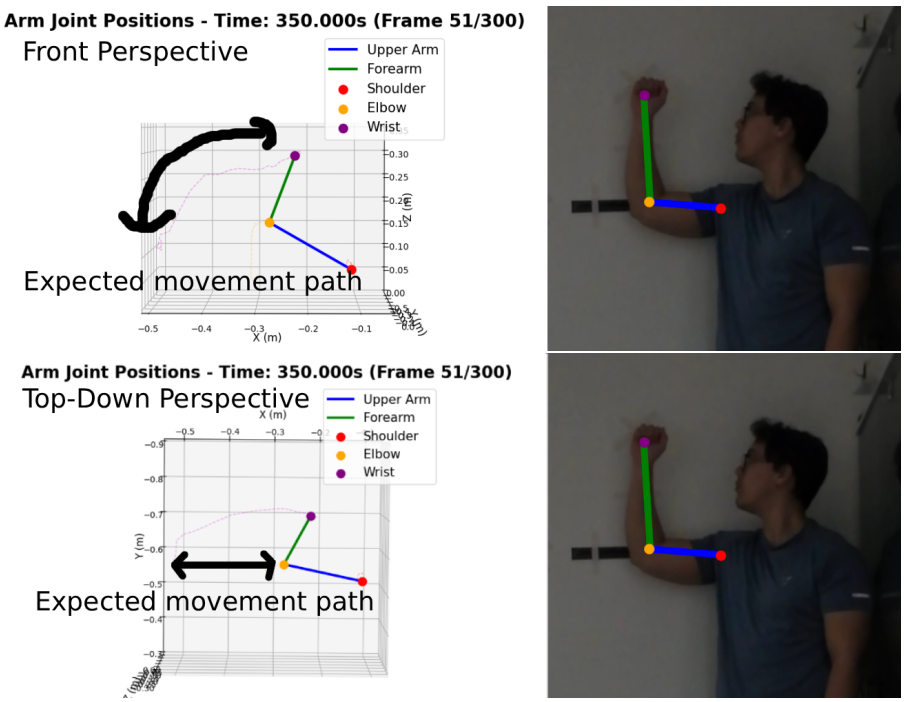


Fig. 4. Front and top view of inferred 3D joint positions.

This path is shown in Fig. 5. This movement has been selected to be easily recreated, whilst showing movement in three different joint angles simultaneously. The same data processing for the two dimensional elbow flex case is used to obtain RMSE of three different joint angle movements.

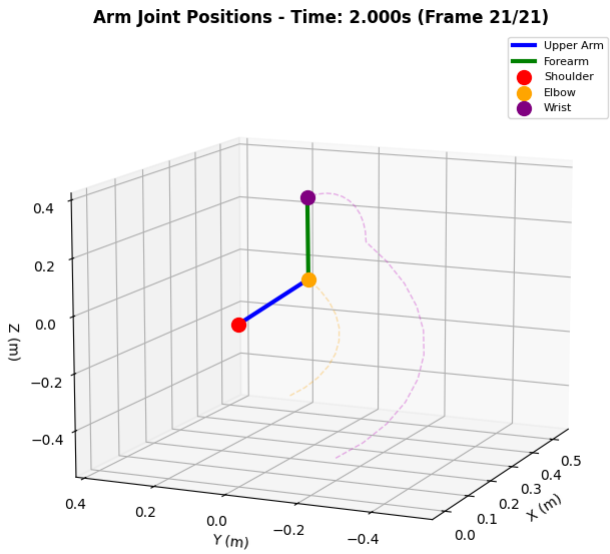


Fig. 5. 3D Joint Motion Path

3.2 Results of joint angle tracking methods

Table 2 shows the compiled RMSE errors for tracking each joint in both the two dimension elbow flexing case and the three dimension movement path.

Table 2. RMSE error of either tracking method.

Method	**RMSE Error Elbow Flexion 2D (°)**	**RMSE Error Elbow Flexion 3D (°)**	**RMSE Error Shoulder Flexion (°)**	**RMSE Error Shoulder Abduction (°)**
Pose estimation	25.83	7.19	6.89	11.68
Marker Tracking	N/A	N/A	N/A	N/A

Whilst the two dimensional RMSE error is larger than the literature suggests, the three dimensional joint error shows better accuracy than expected. Apart from the shoulder adduction/abduction measurements, each joint being tracked is almost in the same order of magnitude of the $3°$ requirements of the project.

Despite various attempts, ArUco marker tracking seemed to be too unstable in this specific application. Both 2D paper markers and 3D printed markers were used, though the glossy surface finish of either meant that tracking was unreliable at close distances and impossible at moderate distances. This was obtained from an experiment wherein a marker is held up to the camera and rotated around various axes, measuring the captured rotation.

Fig. 6 shows a plot of the measured roll angle, and shows the gaps in which no measurement was made. Likewise, the background conditions of the image also made a major difference to whether or not the marker was tracked. For example, Fig. 7 shows the marker not tracking whilst not distorted by angle at all, compared to being tracked at a very large angle. Clearly this level of unreliability is unsuitable for the final tracking task, and therefore no further testing will be required.

An interesting aspect that was encountered when performing the accuracy characterisation experiments with pose estimation is shown in Fig. 8, and points to a downside not immediately apparent. The program uses machine learning to create an estimation of joint locations in an image and was ostensibly trained on a large corpus of human images. Hence, there was trouble distinguishing between real humans and pictures of humans being worn on the subject's clothing - a downside in the favour of the marker tracking method.

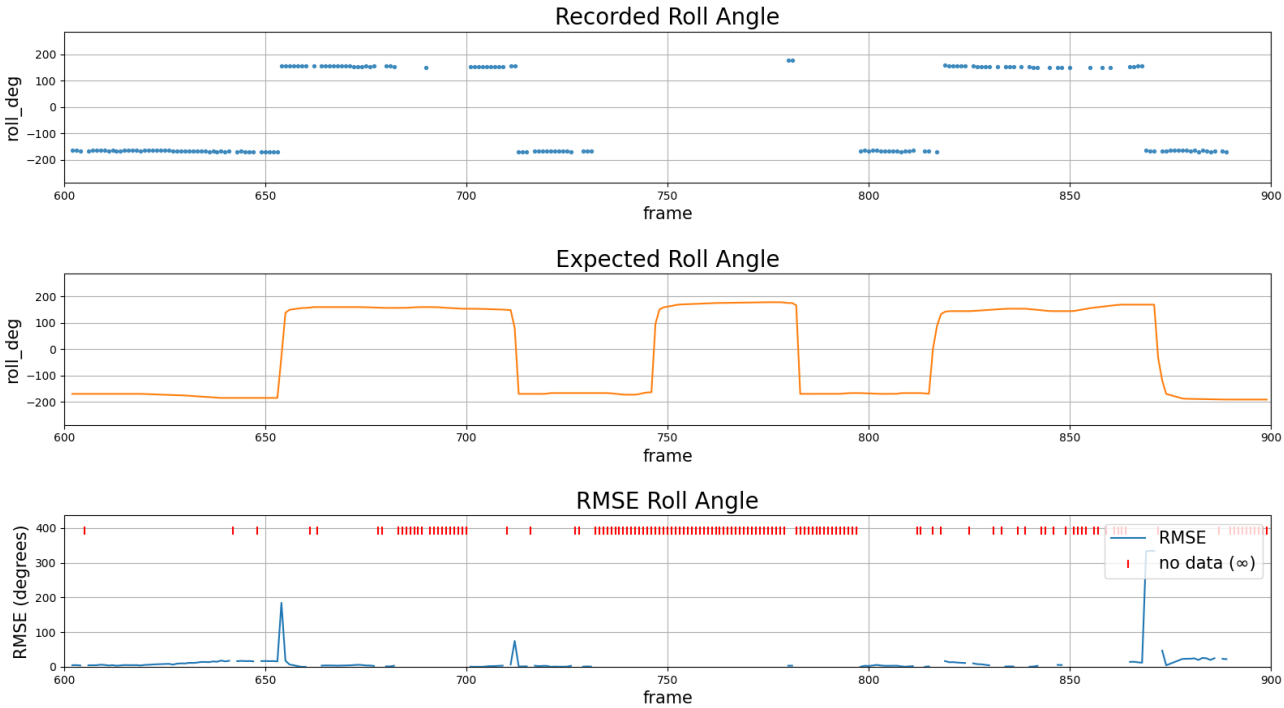


Fig. 6. Measured vs Expected graph of roll angle of ArUco marker.

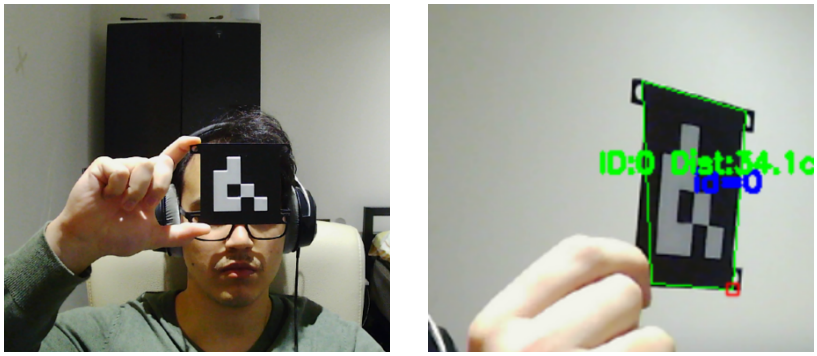


Fig. 7. Background dependence of the ArUco marker.

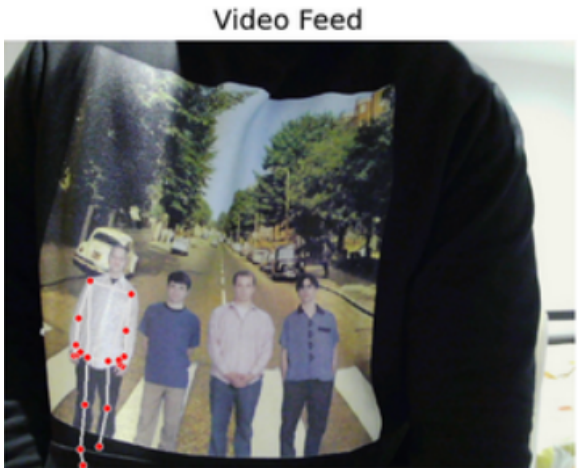


Fig. 8. Pose estimation software mistaking a picture of humans printed on clothing for an actual person.

3.3 Final Tracking Method

Results show that the pose estimation model is not able to achieve the $3°$ required for the project, it is able to get within the same order of magnitude of accuracy. Moreover, the mediapipe model used to power this is optimised for real time measurement; using a 30 fps camera, each measurement takes 33 ms, well below the required 300 ms. Hence, the comparison between conventional joystick control and human arm tracking teleoperation will use the pose estimation approach for comparison.

4 Arm tracking vs conventional control experiments

This section explains three scenarios that have been devised in CoppeliaSim to try and characterise the performance of each tracking method.

4.1 Description of experiment to compare conventional vs human arm tracking

In order to compare fairly the two methods of control, three scenarios will be examined:

Section 4.2 - Position matching task similar to Nisky et al. [14]. A random location within the envelope of the arm is selected and the operator must control the arm to this location as fast as possible.

Section 4.3 - Pick and place task. An object and target location spawns within the envelope of the arm, and the operator must manipulate the object to the target.

Section 4.4 - Obstacle laden pick and place task. Similar to pick and place task, but with tracking of obstacle collisions.

The operator must stand further away from the screen to be in frame of the camera for the pose estimation method. To negate the perception advantages this would incur for the controller tasks, the controller method will be performed sitting this same distance away.

To measure the performance of the position matching task, the distance covered is divided by the time taken to find the normalised speed measurement.

4.2 Scenario 1: Random position matching task

In this scenario a random position within the reachable envelope of the arm is marked and the operator must move the grippers of the arm to within 0.06m of this location as fast as they can. This time is then recorded, and then used to divide the distance where the arm started from to find the average speed of the arm during the experiment.

100 samples were taken, with 10 runs of 10 experiments. Analysis of variance (ANOVA) shows no significant deviation between each set of experiments (see appendix Appendix E for detailed ANOVA results). The controller method was 42.1% faster than the pose estimation algorithm, although with a lower standard deviation of 0.043 compared to 0.053.

The speed difference is likely attributable to the operator's familiarity of using controllers for other tasks. Notably, the difference in standard deviation could imply that pose estimation is stronger for tasks that require consistency, rather than outright speed. However, this standard deviation figure is small enough to be attributable to noise, if nothing else.

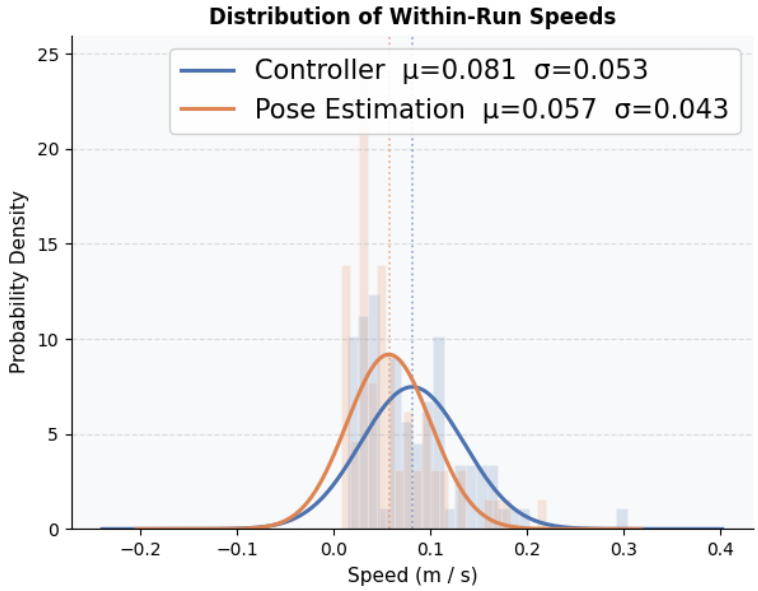


Fig. 9. Standard distribution plot of the results of either experiment.

4.3 Scenario 2: Simple environment pick and place task

Scenarios 2 and 3 consist of four different phases shown in Fig. 10 The object is considered 'held' as long as YuMi's grippers are within a certain distance threshold, and transition from an open state to a closed state. This prevents the experiment from being limited by the design and simulation of a suitable end effector - though this may act as an interesting extension for future research.

To measure the performance of the pick and place task, the cumulative time spent in each phase is measured, along with the overall time to complete the task. Likewise, to ensure fairness between each method in both pick and place tasks, seeded randomness will be used to ensure that both methods get the same trial spawn position.

The pose estimation algorithm shows 24.4% reduction in time over the controller control in this task, with the noticeable difference being in the proportion of time spent in the 'carry' phase (see Fig. 11) The controller method requires the operator to orient the gripper over the object, and doing this in certain arm positions can lead to unreachable positions from the inverse kinematics (IK) solver, hence the longer time.

A graph showing when the IK solver is unable to reach a position is shown in Fig. 12. The advantage of the pose estimation method is that this failure mode is less common, since the joint limitation of the operator's arm generally matches the joint limits of the robot arm.

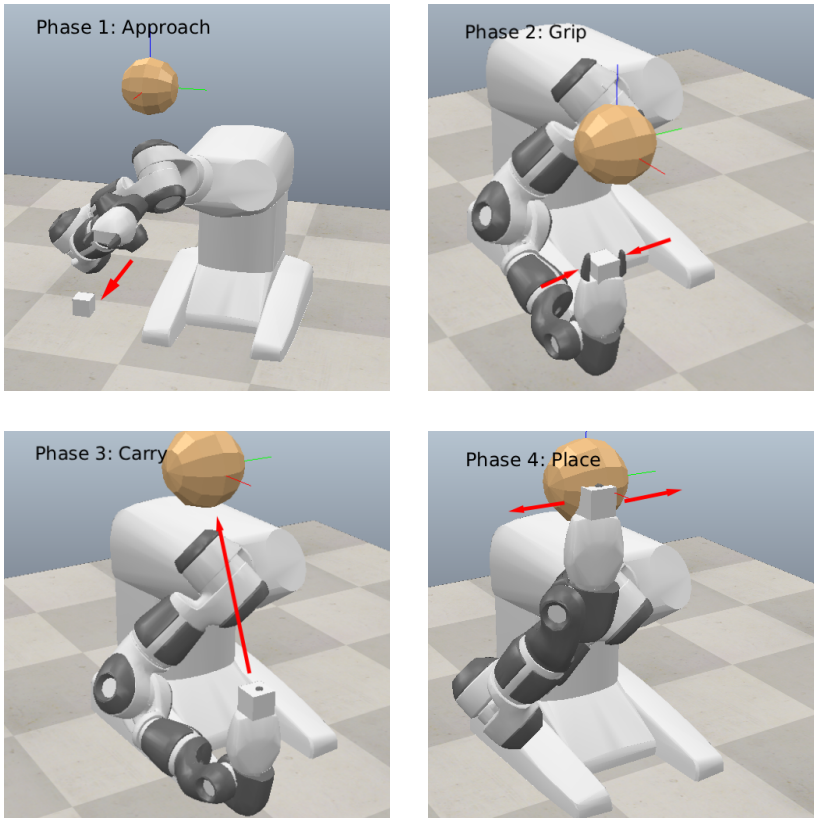


Fig. 10. Phases of the pick and place scenario.

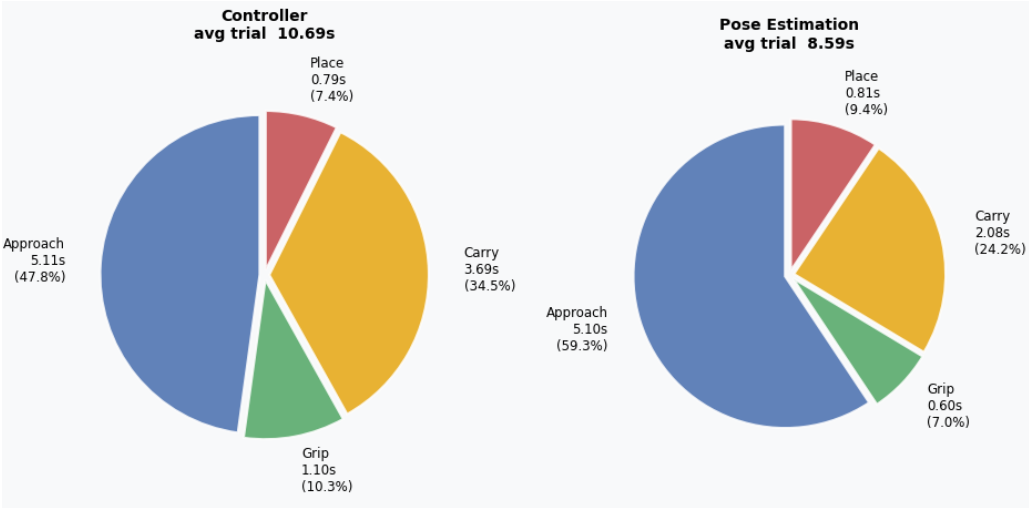


Fig. 11. Average time spent in each phase for either control method in the transport experiment.

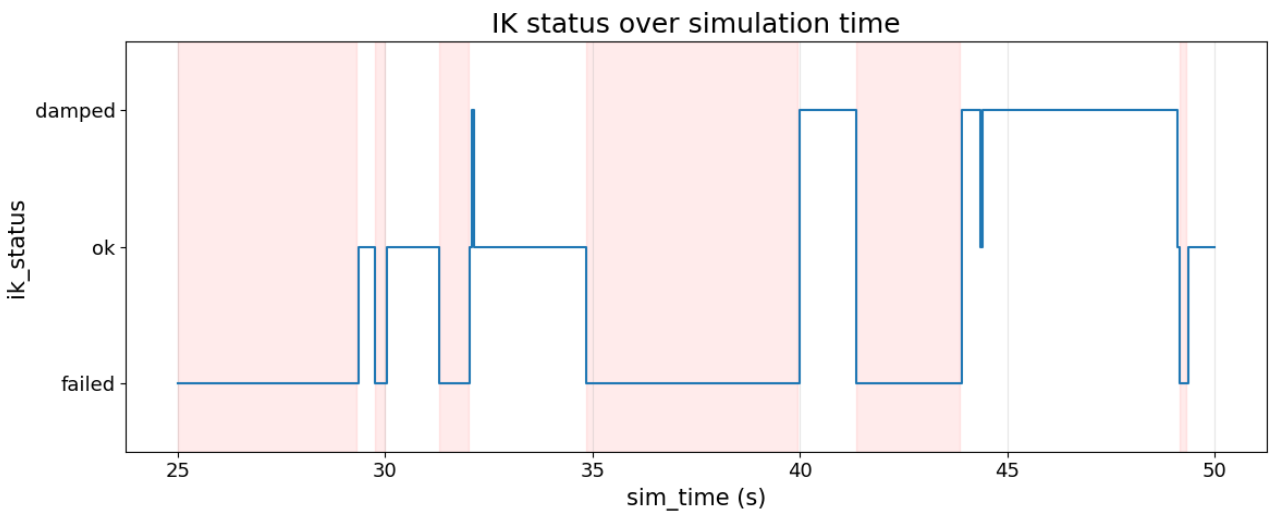


Fig. 12. A snapshot of when the IK solver was failing over the simulation time.

4.4 Scenario 3: Obstacle-rich environment pick and place task

This scenario is the same as scenario 2, with the addition of a randomly generated obstacle field as shown in Fig. 13. The collision between any segment of the robot arm and these obstacles are counted, although there is no time penalty for a collision.

The pose estimation algorithm is quicker on average by 26.7% in this task, likely for the same reasons as above - where the controller solution is running into unsolvable positions. Additionally, the added dexterity shown led to 3.2 fewer collisions (4.4 vs 7.6) on average compared to the controller solution.

The fewer collisions may be due to more control that the operator has over the entire arm, as more data can be transmitted from the operator to the robot, in a way that is intuitive. Likewise, the smooth

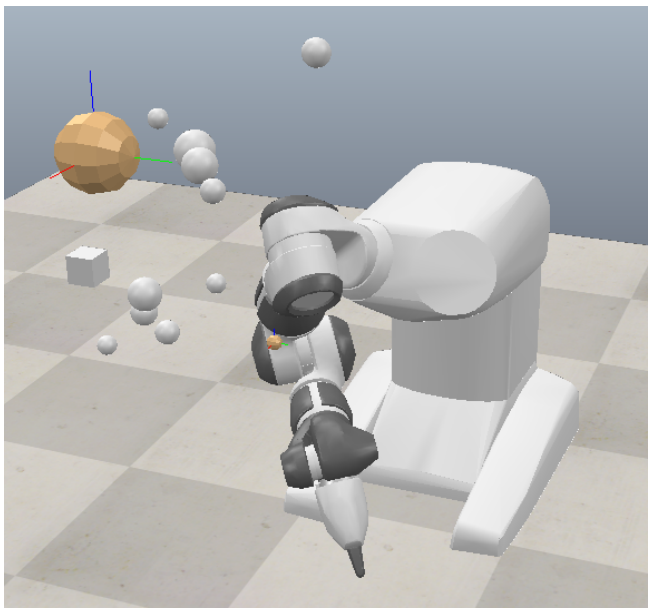


Fig. 13. One of the obstacle fields generated for scenario 3.

control of the pose estimation algorithm may make it more predictable compared to the IK approach of the controller solution - where it may be jumping over singularity positions unpredictably.

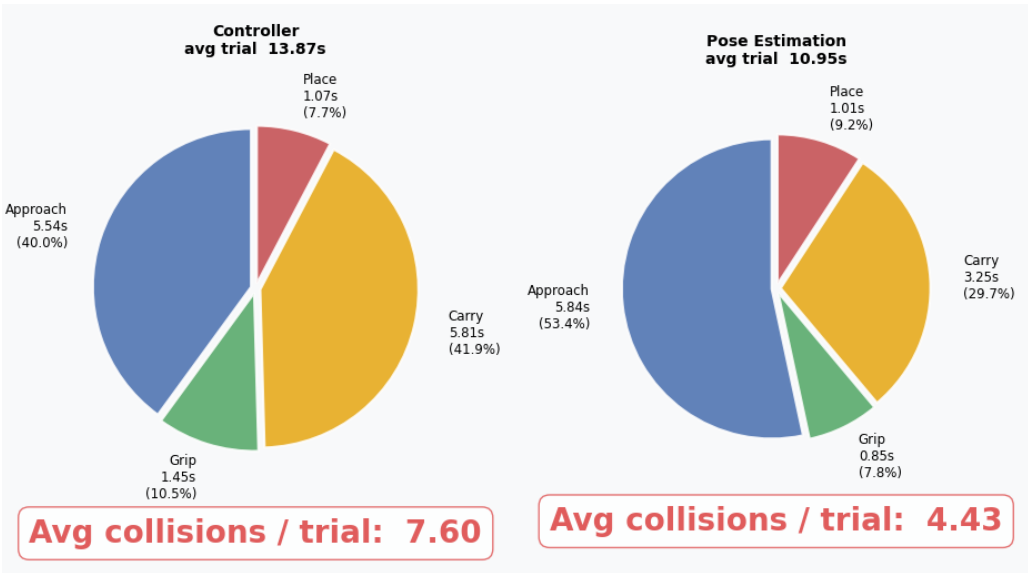


Fig. 14. Standard distribution plot of the results of either experiment.

5 Conclusions, future work and reflections

This section summarises the major findings from the experiments carried out in the previous section, and provide several areas to be explored in future work.

5.1 Conclusions

The results of the experiments shows that in the arbitrary movement task, joystick control results in a 42.1% faster average speed to reach a random position. This could have been caused by multiple factors: the operator's proficiency with using joysticks previously, as well as intuitively the joystick control being more well suited to this task. It follows that the joystick is able to manipulate the end effector coordinates one axis at a time to reach the target position systematically, whereas it is harder to move a hand to an arbitrary position.

However, the use of pose estimation had increased performance in the two pick and place scenarios, being faster on average by 24.44% and 26.67% respectively. This is likely a combination of human intuition being more applicable, as well as a limitation of the inverse kinematics solver used for the joystick control. Simply, the pose estimation solution is able to deliver 7 degrees of control to the arm simultaneously and intuitively, compared to the abstraction required to manipulate the arm with a joystick.

5.2 Future work

There are many ways to improve on the scope of this project, and the fidelity of the results. The simplest would be for a larger sample size of operators - due to the limited time and budget of this project only the author was able to participate in data gathering for the experiments. Using a wide range of participants, and noting their level of proficiency with using joysticks, could inform any correlation between performance increase and skill.

Likewise, the pose estimation method is not necessarily the best representation of true teleoperation as discussed by Jankowski and Grabowski [6]. This is because the experiments show a third person perspective of the robots workspace, and not a point of view (POV) camera where the head may be. Hence, there is still a level of control abstraction as the operator has to translate between coordinate spaces (which admittedly took very little time to get used to). Future analysis could explore virtual reality (VR) and augmented reality (AR) set ups in conjunction with full arm tracking.

Additionally, as mentioned in Section 4, the end effector is assumed to be a fully efficacious instrument - while wonderful for the purposes of science, not necessarily practical in reality. Future work could expand with different types of end effector designs, and adding further tracking of individual fingers to be mapped to a manner of end effector designs.

5.3 Project Management Reflections

The author's previous placement experience as a software engineer (specialising in Python and networked microservices) made many of the technical aspects of this project achievable. For example, the code to control simulation environment used the same publisher / subscriber model made it possible to progress from basic robot control to full teleoperation code very quickly. Likewise, all of the data gathering and processing for the experiments was done with bespoke Python scripts, allowing for swift iteration and progress.

Combined with the author's strict adherence to deadlines and planned milestones (and a purposefully strict project plan), allowed for a thorough project to be completed. It would be unfair to not mention the supervisor, Dr. Watson, who aided in furthering of many of the ideas in this project, and providing the motivation to finish the project.

References

[1] D. Nguyen Thai et al., "Biomechanical analysis of the shooter-weapon system oscillation," *2017 International Conference on Military Technologies (ICMT)*, pp. 48–53, May 2017. DOI: 10.1109/miltechs.2017.7988729 (cited on p. 8).

[2] H. Zhou et al., "Human–robot teleoperation based on imus," *Advanced Topics in Science and Technology in China*, pp. 51–72, 2025. DOI: 10.1007/978-981-96-6139-8_3 (cited on pp. 8, 10).

[3] H. Lyu, G. Yang, and H. Yang, "Whole-body human-motion based robot teleoperation," *Advanced Topics in Science and Technology in China*, pp. 69–93, 2025. DOI: 10.1007/978-981-96-6545-7_4 (cited on p. 8).

[4] D. Stanev and K. Moustakas, "Modeling musculoskeletal kinematic and dynamic redundancy using null space projection," *PLOS ONE*, vol. 14, no. 1, Jan. 2019. DOI: 10.1371/journal.pone.0209171 (cited on p. 8).

[5] H. Jin, H. Ji, and F. Yan, "An effective obstacle avoidance and motion planning design for underwater telescopic arm robots based on a tent chaotic dung beetle algorithm," *Electronics*, vol. 12, no. 19, p. 4128, Oct. 2023. DOI: 10.3390/electronics12194128 (cited on p. 8).

[6] J. Jankowski and A. Grabowski, “Usability evaluation of vr interface for mobile robot teleoperation,” *International Journal of Human-Computer Interaction*, vol. 31, no. 12, pp. 882–889, Nov. 2015. DOI: 10.1080/10447318.2015.1039909 (cited on pp. 8–10, 23).

[7] E. B. Mazomenos et al., “Detecting elementary arm movements by tracking upper limb joint angles with marg sensors,” *IEEE Journal of Biomedical and Health Informatics*, vol. 20, no. 4, pp. 1088–1099, Jul. 2016. DOI: 10.1109/jbhi.2015.2431472 (cited on p. 9).

[8] L. Phuong and V. Cong, “Control the robot arm through vision-based human hand tracking,” *FME Transactions*, vol. 52, no. 1, pp. 37–44, 2024. DOI: 10.5937/fme2401037p (cited on p. 9).

[9] V. Nalam and P. Manivannan, “Development of a contact based human arm motion analysis system for virtual reality applications,” *Applied Mechanics and Materials*, vol. 592–594, pp. 2139–2144, Jul. 2014. DOI: 10.4028/www.scientific.net/amm.592-594.2139 (cited on p. 9).

[10] X. Liu et al., “Video based human motion capture,” *1999 IEEE Third Workshop on Multimedia Signal Processing (Cat. No.99TH8451)*, pp. 483–488, 1999. DOI: 10.1109/mmsp.1999.793896 (cited on p. 9).

[11] B. Milling, *Motion capture: Why is it needed, and how does it work?* Jul. 2024. [Online]. Available: https://www.americanmovieco.com/blog/motion-capture-why-is-it-needed-and-how-does-it-work (cited on p. 10).

[12] M. Crawford, *What is motion capture (mo-cap): The definitive guide*, Mar. 2025. [Online]. Available: https://filmlifestyle.com/what-is-motion-capture/ (cited on p. 10).

[13] G. Doisy, A. Ronen, and Y. Edan, “Comparison of three different techniques for camera and motion control of a teleoperated robot,” *Applied Ergonomics*, vol. 58, pp. 527–534, Jan. 2017. DOI: 10.1016/j.apergo.2016.05.001 (cited on p. 10).

[14] I. Nisky, M. H. Hsieh, and A. M. Okamura, “Uncontrolled manifold analysis of arm joint angle variability during robotic teleoperation and freehand movement of surgeons and novices,” *IEEE Transactions on Biomedical Engineering*, vol. 61, no. 12, pp. 2869–2881, Dec. 2014. DOI: 10.1109/tbme.2014.2332359 (cited on pp. 10, 18).

[15] J. Louca et al., “Impact of haptic feedback in high latency teleoperation for space applications,” *ACM Transactions on Human-Robot Interaction*, vol. 13, no. 2, pp. 1–21, Jun. 2024. DOI: 10.1145/3651993 (cited on p. 10).

[16] B. Wieczorek et al., “The effects of aruco marker velocity and size on motion capture detection and accuracy in the context of human body kinematics analysis,” *Technical Transactions*, pp. 1–10, 2020. DOI: 10.37705/techtrans/e2020036 (cited on p. 12).

[17] H. Wang et al., “A computer-vision method to estimate joint angles and l5/s1 moments during lifting tasks through a single camera,” *Journal of Biomechanics*, vol. 129, p. 110 860, Dec. 2021. DOI: 10.1016/j.jbiomech.2021.110860 (cited on pp. 12, 14).

[18] T. Starner et al., “The perceptive workbench: Computer-vision-based gesture tracking, object tracking, and 3d reconstruction for augmented desks,” *Machine Vision and Applications*, vol. 14, no. 1, pp. 59–71, Apr. 2003. DOI: 10.1007/s00138-002-0096-8 (cited on p. 12).

[19] L. Ricci, F. Taffoni, and D. Formica, “On the orientation error of imu: Investigating static and dynamic accuracy targeting human motion,” *PLOS ONE*, vol. 11, no. 9, Sep. 2016. DOI: 10.1371/journal.pone.0161940 (cited on p. 12).

[20] T. InvenSense, *Icm-20948 datasheet*, 2017 (cited on p. 12).

[21] G. Saggio, “Bend sensor arrays for hand movement tracking in biomedical systems,” *2011 4th IEEE International Workshop on Advances in Sensors and Interfaces (IWASI)*, pp. 51–54, Jun. 2011. DOI: 10.1109/iwasi.2011.6004685 (cited on p. 12).

Appendices

A Project outline

A.1 Overview and motivations

The goal of this project is to create a system for controlling a laterally mounted robot arm with a human arm. Ideally, the system will map each degree of freedom (DoF) of the human arm to the robot arm.

The motivation for this project are the two books: Revitalizing Human-Robot Interaction (Zhou et al) and Human Robot Awareness and Robot Teleoperation (Honghao et al), which describes research into remote teleoperation of laterally mounted robot arms. These books explain the use of an inertial measurement unit (IMU) device which maps the movement of human arms to robot arms, however they only map the end effector position to the position of the wrist.

My project would aim to create a device allowing for mapping of the full robot arm to an operator's arm. In an obstacle-rich environment, mapping of the end effector position is insufficient. The main advantage of teleoperation is to allow the human operator to make real time judgments for complex situations, and thus fully mapping out the operator's intentions to the robot is crucial.

One such scenario that is researched is given in the paper by Jin et al, describing a complex obstacle avoidance algorithm for moving a robot arm underwater for sample retrieval. The drawback mentioned in this paper is that both the sample and arm origin is stationary, and the algorithm takes up to ten seconds to solve this simple case. Ostensibly, one can imagine the complexity required for a scenario where both the sample, arm, and obstacles are moving - solving this autonomously is inefficient compared to using human intuition.

A.2 Deliverables

The deliverables for this project are as follows:

- The creation of a device or software that is capable of reading the angles in an operator's arm, in real time.

- Part of the project scope will be the rigorous examination of existing methods, and justifying the chosen method.
- Applying the values read in from this software to a simulation environment, where a laterally mounted 7 DoF robot arm is controlled from the operator's arm movements.
- Creating an obstacle-rich simulation, where an operator is tasked with moving between points to quantify the differences in success criteria over conventional control methods (e.g. a joystick or traditional end effector mapping).
- Creation of a final report that outlines a table of contents, introduction, methodology, results and conclusion.

B Risk assessment

MANCHESTER 1824
The University of Manchester

RISK REGISTER FOR 3rd YEAR PROJECT

Project Title:	Human Arm Tracking Human Robot Interface		Submission Date:	17/10/25	
Student Name:	Nicholas	Teague			

Project Risk	Severity			Potential			Score (Severity x Potential)	Mitigation Measures
	L	M	H	L	M	H	L=1, M=2, H=3	
Repetitive Strain Injuries from robot operation		x			x		4	Limit testing sessions to less than 30 minutes, ensuring that the operator is adequately warmed up with stretches. Pay attention to muscle discomfort and fatigue, immediately stopping all activity if experienced.
Muscle strains from attempted movements beyond joint capabilities		x			x		4	Program software to show operator limits for joint positions, design control mapping to stay comfortably within human movement ranges.
Eye strain and postural fatigue			x		x		6	Take regular breaks from seated work, e.g. walking around the room, focusing eyes on a far away point, etc.
Mental exhaustion		x				x	6	Schedule regular breaks from work, and avoid working at obscene hours in the day. Vary work tasks, and try to maintain a sufficient work-life balance and sleep schedule.
Motion sickness		x		x			2	Minimise system latency, immediately stop tests if discomfort occurs. Avoid extended testing situations.
Collision with objects in workspace	x					x	3	Keep work area, especially areas in the workspace of the arm clutter-free.

C Code

All relevant code (and source for this report) can be found on my GitHub @ https://github.com/coding-beagle/UoMFinalProject.git

While at it, please check out https://www.nteague.com

D DH Parameters

The DH parameters used for the arm are as follows:

Table 3. Denavit-Hartenberg Parameters for 7-DoF Arm (where L_u is upper arm length and L_l is lower arm length)

Joint	Description	θ_i	α_i	d_i	a_i
1	Shoulder abduction/adduction	q_1	$\frac{\pi}{2}$	0	0
2	Shoulder flexion/extension	q_2	$\frac{\pi}{2}$	0	0
3	Shoulder rotation	q_3	$-\frac{\pi}{2}$	0	L_u
4	Elbow flexion/extension	q_4	$\frac{\pi}{2}$	0	0
5	Forearm rotation	q_5	$-\frac{\pi}{2}$	0	L_l
6	Wrist flexion/extension	q_6	$\frac{\pi}{2}$	0	0
7	Wrist radial/ulnar deviation	q_7	0	0	0

E ANOVA Figures

```
Running ANOVA analysis for pose estimation results:

Pose Estimation - loading 10 file(s):
Good! Group A: reach_results_20260301_232316  [reach]  (n=8  =9.769s)
```

```
Good! Group A: reach_results_20260301_234021  [reach]  (n=7   =5.787s)
Good! Group A: reach_results_20260301_235136  [reach]  (n=7   =10.475s)
Good! Group A: reach_results_20260301_235234  [reach]  (n=9   =2.589s)
Good! Group A: reach_results_20260301_235402  [reach]  (n=10   =7.558s)
Good! Group A: reach_results_20260301_235622  [reach]  (n=8   =9.298s)
Good! Group A: reach_results_20260307_014934  [reach]  (n=6   =5.696s)
Good! Group A: reach_results_20260307_015112  [reach]  (n=8   =5.325s)
Good! Group A: reach_results_20260307_015228  [reach]  (n=10   =6.336s)
Good! Group A: reach_results_20260307_015437  [reach]  (n=7   =7.959s)
```

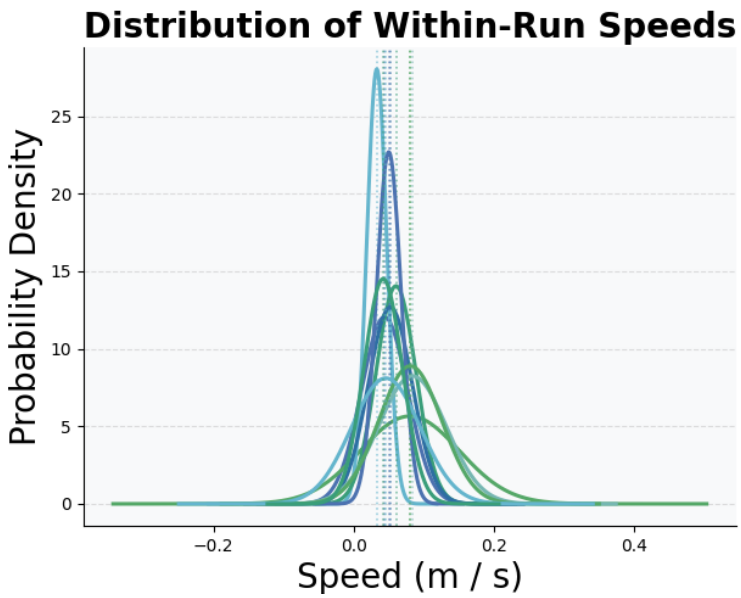


Fig. 15. Compound standard distribution plot of all 10 pose estimation runs.

```
Running ANOVA analysis for controller results:

Controller - loading 10 file(s):
Good! Group A: reach_results_20260307_003628  [reach]  (n=10   =6.077s)
Good! Group A: reach_results_20260307_003835  [reach]  (n=8   =8.768s)
Good! Group A: reach_results_20260307_004001  [reach]  (n=8   =2.839s)
Good! Group A: reach_results_20260307_004142  [reach]  (n=9   =7.128s)
Good! Group A: reach_results_20260307_004240  [reach]  (n=10   =3.770s)
Good! Group A: reach_results_20260307_011713  [reach]  (n=8   =7.147s)
Good! Group A: reach_results_20260307_011850  [reach]  (n=9   =6.670s)
```

```
Good! Group A: reach_results_20260307_011956  [reach]  (n=10   =4.739s)
Good! Group A: reach_results_20260307_012253  [reach]  (n=8   =7.331s)
Good! Group A: reach_results_20260307_012436  [reach]  (n=10   =8.853s)
```

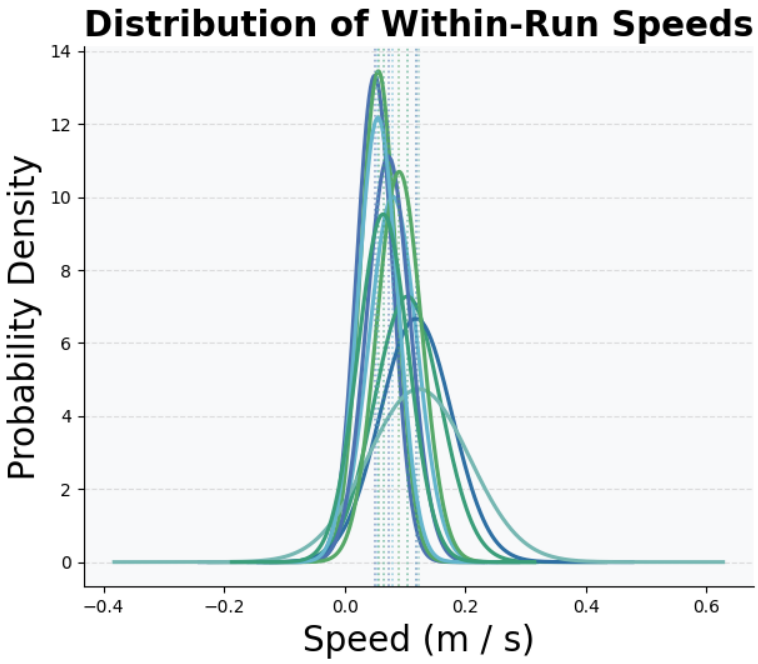


Fig. 16. Compound standard distribution plot of all 10 controller runs.